$$y = \frac{x-a}{x-b}e^x$$

$$\begin{aligned}\frac{dy}{dx} &= e^x\left(\frac{x-a}{x-b} + \frac{a-b}{(x-b)^2}\right)\\ &= \frac{e^x}{(x-b)^2}\big((x-a)(x-b)+a-b\big)\\ &= \frac{e^x}{(x-b)^2}\big(x^2-(a+b)x+ab+a-b\big)\end{aligned}$$

$$\begin{aligned}\Delta > 0 \implies \big(a+b\big)^2 - 4(ab+a-b) &> 0\\ (a-b)(a-b-4) &> 0\end{aligned}$$

$$a-b<0 \quad \text{or} \quad a-b>4$$

(i)

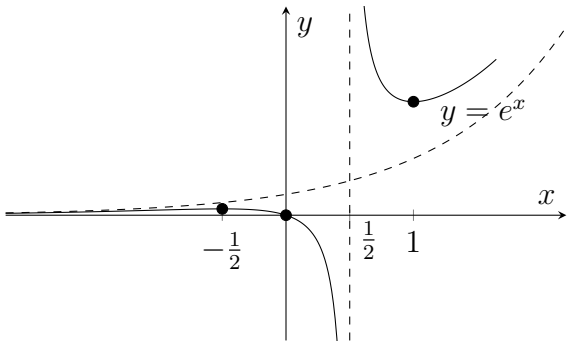
$$\begin{aligned}x^2 - \frac{1}{2}x - \frac{1}{2} &= 0\\ 2x^2 - x - 1 &= 0\\ (2x+1)(x-1) &= 0\end{aligned}$$

$$x = -\frac{1}{2}, \quad x = 1$$

Vertical asymptote: $x = \dfrac{1}{2}$

$$\therefore -\frac{1}{2} < \frac{1}{2} < 1$$

x	$-\infty$	$-\frac{1}{2}$	0	$\frac{1}{2}^-$	$\frac{1}{2}^+$	1	∞
y	0	$\frac{e^{-1/2}}{2}$	0	$-\infty$	∞	$2e$	∞



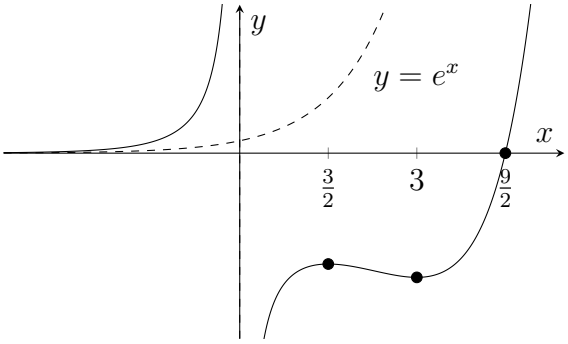
(ii)

$$x^2 - \frac{9}{2}x + \frac{9}{2} = 0 \implies (2x-3)(x-3) = 0$$

$$x = \frac{3}{2}, \qquad x = 3$$

Vertical asymptote: $x = 0$

x	$-\infty$	0^-	0^+	$\frac{3}{2}$	3	∞
y	0	∞	$-\infty$	$-\frac{3}{2}e^{3/2}$	$-\frac{1}{2}e^3$	∞



This solution was submitted by **Jon**

Published and refined by **Jon**

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